

HOTEL MANAGEMENT SYSTEM

Comprehensive Project Scope & Database Design Report

1. Introduction and Project Objective

The present project has been developed with the goal of designing and implementing a web-based **Hotel Management System (HMS)**. This system is designed to enable hotel managers, staff, and guests to manage reservation, accommodation, and service processes in a simple, organized, and smart manner. Its main objective is to facilitate daily hotel operations, including registering reservations, managing payments, checking room statuses, and receiving customer feedback.

The overall workflow of the hotel management system is designed as a cohesive cycle between users (guests) and managers (hotel staff) so that all processes—from the search stage to room checkout—are performed automatically and intelligently.

In this system, after logging in, the user (guest) can view the list of available rooms and select their desired option using filters such as room type, capacity, time period, and price. Once a room is selected, the initial reservation process begins, and required information such as check-in and check-out dates, number of individuals, and the required type of services is registered. Subsequently, the user proceeds to the electronic payment stage and completes the payment through a banking gateway. Upon payment confirmation, the system automatically issues an electronic invoice and a unique reservation ID, assigning them to the user's account.

During the stay, the user has access to features such as requesting additional services (laundry, room cleaning, food or beverage service, extension of stay, etc.) and can communicate with the reception desk through their user panel. The system also allows the user to view and track the status of their requests. At the end of the stay, the settlement and checkout processes are performed, and the guest can register their reviews, suggestions, and satisfaction levels through the feedback section. This feedback is stored in the database and utilized by managers to analyze service quality.

On the other hand, staff and system administrators access dedicated modules via the management panel. These modules include managing room information, controlling reservation statuses, registering and reviewing payments, organizing staff work shifts, and generating managerial and analytical reports on the daily or monthly performance of the hotel. Thus, the system administrator is able to stay informed about room occupancy status, revenues, and customer feedback in real-time, enabling them to make more precise managerial decisions.

From a technical perspective, this project is a practical example of designing and implementing a multi-table **Relational Database System** where One-to-Many and Many-to-Many relationships are fully utilized. For instance, each user can have multiple reservations (One-to-Many relationship), while each room may also be reserved by different users on different dates (Many-to-Many relationship through an intermediary table). In addition to demonstrating the concepts of logical and physical database design, this data structure enables

secure data management, reduces data redundancy, prevents human errors, and ultimately increases system efficiency and accuracy. Furthermore, through consistent input and output forms, the user interaction process with the system is designed to be simple and user-friendly.

2. Introduction and Description of Entities and Their Attributes

The system is built upon a set of interconnected **Entities**, each playing a pivotal role in the dynamic operation of the system and communicating with one another through logical and structured relationships. The main entities defined in this project are as follows:

Hotel

The primary and parent entity in the system hierarchy, containing the foundational and identifying information of each hotel.

- **Attributes:** Hotel name, hotel identification code, full address, main contact number, official email, total number of floors, total number of rooms, quality rating (e.g., number of stars), general description of public amenities (such as swimming pool, conference hall, sports complex, parking), name of the CEO or main manager, and official website.

Customer (Guest)

The primary users of the system by whom reservation operations, service usage, and feedback registration are performed.

- **Attributes:** First name, last name, national ID, birth certificate number, date of birth, gender, mobile phone number, landline phone number, email address, residential address, date of first reservation, total number of stays, membership level (Standard, Silver, Gold), membership date, and profile picture (optional).

Room

The physical representative of each accommodation unit in the hotel.

- **Attributes:** Room number, floor number, current room status (vacant, reserved, occupied, under cleaning, out of service), standard capacity, maximum capacity, room type (reference to Room Type), standard nightly price, special descriptions of in-room amenities (such as view, air conditioning system).

Room Type

The logical classification of rooms based on features and amenities to facilitate selection and pricing.

- **Attributes:** Room type name (such as Standard, Double, Twin, Junior Suite, Royal Suite), detailed description of the room type and its unique amenities, default capacity, nightly price range, minimum and maximum number of reservations within a specific time period.

Reservation

The official registration of a guest's accommodation request for a specific time period.

- **Attributes:** Unique reservation number, date and time of reservation registration, check-in date, check-out date, number of nights of stay, total number of individuals (adults, children), reserved room ID, reserving customer ID, reservation status (confirmed, pending confirmation, canceled, completed), estimated total reservation amount, registration method (online, phone, in-person), applied discount codes.

Payment

Registration of financial information related to each transaction.

- **Attributes:** Payment identification number, amount paid, exact date and time of payment, payment method (online via banking gateway, card-to-card, cash on-site, bank transfer), transaction tracking number (if online), associated reservation ID, associated invoice ID, payment status (successful, unsuccessful, pending confirmation), origin account name or cardholder name.

Invoice

The final financial document for each stay or set of services.

- **Attributes:** Unique invoice number, invoice issue date, associated reservation ID, customer ID, total payable amount, amount paid, balance due, detailed breakdown of costs (including accommodation cost, additional service costs, value-added tax, discounts), name of the issuing staff member, final invoice status (settled, unsettled).

Service (Hotel Services)

Registration and management of requests for ancillary services.

- **Attributes:** Service ID, service name (such as room service, laundry, taxi service, massage, café), service description, base tariff per service unit, unit of measurement (hourly, quantity, kilogram), service delivery time window, name of the staff member responsible for service delivery, room ID (if specific), requesting customer ID.

Staff (Hotel Employees)

Management of information regarding personnel employed at the hotel.

- **Attributes:** Personnel code, first name, last name, national ID, date of birth, contact number, residential address, job title, system access level (manager, reception, cleaning supervisor, cashier), date of hire, base salary, banking information (for salary deposit), employment status (active, on leave, terminated).

Feedback

Collection of reviews, suggestions, and criticisms from guests.

- **Attributes:** Feedback ID, associated reservation ID, customer ID, feedback registration date, feedback text, overall rating from 1 to 5, room quality rating, service quality rating, staff behavior rating, feedback handling status (pending review, under investigation, completed).

3. Detailed Description of the Overall Project Workflow and Entity Relationships

The hotel management system is a web-based software designed based on a three-tier architecture (Frontend, Backend, Database). The overall workflow of the project is that the user (guest) first logs into the customer portal and views their user space. In this space, the option to view details of various room types, check vacant capacity for desired dates, and compare amenities is provided. After selection, the customer can register their reservation. This reservation is considered a "pro-forma invoice" and its status remains "pending confirmation" until payment is made.

Once the reservation is registered, the system stores the relevant information in the **Reservation** entity. Then, the customer is directed to the payment page, where final amount details are displayed and the payment method can be selected. Upon successful completion of the payment, the system registers it in the **Payment** entity and changes the reservation status to "confirmed". This financial transaction will form the basis for issuing the final **Invoice**, which includes all costs of accommodation and requested services.

During the stay, the customer can request additional services, such as room service or laundry, through their user panel. These requests are registered in the **Service** entity, and the responsible staff member can view and execute them from their own panel. The cost of these services will be added to the final invoice.

Hotel staff manage their assigned duties through their dedicated panel. Reception is responsible for confirming reservations and handling guest check-ins and check-outs. The cleaning team updates room readiness statuses. Service department staff receive and execute customer requests. Hotel managers have full access to all sections, including managing hotel entities and staff, viewing all reservations and payments, and most importantly, receiving various analytical and managerial reports on the daily or monthly performance of the hotel.

Access levels are designed hierarchically:

- **Admin (Manager):** Full access to all features, including viewing, adding, editing, and deleting all entities (hotel, customer, room, reservation, payment, staff, service, feedback, invoice).
- **Staff (Employee):** Access to a subset of entities depending on their job role; for example, reception can view and edit reservations and customer information but cannot alter hotel details. Cashiers have access to the payment and invoice sections, and the cleaning supervisor updates room statuses.
- **Customer (Guest):** Limited access to information and operations related to themselves: viewing and registering reservations, making payments, viewing invoices, requesting services, and registering feedback.

Relationships between entities are defined logically based on the database structure:

- A hotel can have **multiple** rooms and **multiple** staff members (One-to-Many relationship).
- A customer can make **multiple** reservations, and each reservation is registered by **one** customer (One-to-Many relationship).
- Each reservation belongs to **one** specific room and **one** specific room type (One-to-One or One-to-Many relationship depending on the design).
- Each reservation can lead to **multiple** payments (in case of installment payments), but each payment is typically associated with **one** reservation (One-to-Many relationship).
- Each reservation and its set of payments ultimately generate **one** invoice (One-to-One relationship).
- A staff member can provide **multiple** services, and each service is performed by **one** staff member (One-to-Many relationship).
- Feedbacks belong to **one** specific reservation (One-to-One or One-to-Many relationship).

These relationships enable complex data queries and comprehensive reporting.

4. Sample Reports and Outputs Extractable from the System

Below are 10 key examples of reports and outputs that the hotel management system is capable of producing:

1. **Active and Upcoming Reservations Report:** Displays a list of all reservations whose check-in date is from today onward, along with the guest's name, room number, check-in and check-out dates, and payment status. This report is vital for the daily planning of the reception desk.
2. **Currently Resident Guests Report:** A list of all guests currently staying in the hotel, including name, room number, expected check-in and check-out dates, and any special service requests.
3. **Occupied and Vacant Rooms Report:** Displays the real-time status of all hotel rooms, categorized into "vacant", "reserved", "under cleaning", "occupied", and "out of service". This feature helps the housekeeping and reception departments achieve better management.
4. **List of Customers with Luxury Rooms Reserved:** A list exclusively consisting of guests who have reserved "luxury" or "suite" room types, along with their reservation details and contact information. This is useful for providing special services or targeted marketing campaigns.
5. **Daily/Weekly/Monthly Revenue Report:** A summary of all successful payments registered within a specific time period, broken down by revenue type (accommodation, services, room service, etc.) and payment method. This is a key financial report for management.
6. **List of Employees and Their Job Titles:** Displays complete information of all hotel personnel, including first name, last name, personnel code, job title, system access level, and date of hire. This is essential for human resource management.
7. **Guests' Requested Services Report:** A list of all additional services requested by guests, including service name, guest name, room number, request date and time, and the staff member responsible for execution. This helps evaluate the performance of the service department.
8. **Customer Feedback Analysis:** Displays a summary of reviews and ratings registered by guests, with the ability to filter by date, rating, or feedback type (complaint, suggestion). The overall hotel average rating and ratings for individual sections (room, service, staff) are also viewable.
9. **Unsettled Invoices Report:** A list of all invoices that have not yet been fully settled, including guest name, reservation number, remaining balance, and due date. This report is critical for tracking financial claims.
10. **Complete Stay History of a Specific Guest:** By searching for a customer's name or ID, the system displays all reservations, stays, payments, and feedback registered by them. This information is highly valuable for understanding customer behavior and providing personalized services.